

```
cd:41.70:30:48.07:16:81:e5:30:34:66:f1:56:ef:db
Please contact your system administrator.
Add correct host key in /home/slynnux/.ssh/known_hosts to get rid of this
message.
Offending key in /home/slynnux/.ssh/known_hosts:24
RSA host key for localhost has changed and you have requested strict checking
Host key verification failed.
```

If we're certain about the key identification chance of the remote machine, we can remove the corresponding server key entry from our `~/.ssh/known_hosts` file. Following which, the next time you try to log in, you will be asked for a key verification again and the server key will be again registered in the `known_hosts` file.

Executing remote commands

The main purpose of SSH is to execute commands remotely. As we have already seen, immediately after a successful SSH log-in, we're provided with the remote user's shell prompt from where we can execute all sorts of commands that the remote user is allowed to use. This 'pseudo' terminal session will exist as long as you're logged in.

It is also possible to execute commands on a one-at-a-time basis without assigning a pseudo-terminal, as follows:

```
slynnux@gnubox:~$ ssh slynnux-laptop 'uname -a'
slynnux@slynnux-laptop:~$ ssh slynnux-laptop 'cat /etc/passwd | grep root'
Linux slynnux-laptop 2.6.28-9-generic #31-Ubuntu SMP Wed Mar 11 15:43:58 UTC
2009 i686 GNU/Linux
slynnux@gnubox:~$
```

Note that we're back at our local shell prompt. The syntax is `ssh user@hostname 'commands in quote'`.

Input/output redirection

Piping is a nifty feature provided by the shell. If you aren't already familiar with it, have a look at the basics of piping in the following section.

Piping is used for input and output redirection. In *nix shells, we can redirect input/output in different ways, as follows:

```
echo "Test" > file
```

Here the output text stream ("Test") is directed to a file. Thus the stream is stored to a file named file. '>' is the output redirection operator.

Now, take a look at the following command:

```
cat < file
```

Here, input is directed to the `cat` command. `cat` performs the concatenation of the input stream. Here the input is a file named `file`. '<' is an input redirection operator that directs the input stream to the specified command. Here it directs the input text stream from the file to the `cat`.

Finally, take a look at the following command:

```
echo hello | command1 | command2
```

Here, '|' is the piping operator. It uses the output of one command as the input of another. We can use any number of pipes serially, i.e., the output of one command appears as the input of another, and the output of *this second* command appears as the input of the third command and so on. Thus, the net result will be a serial application of these commands on data, one after the other.

For example:

```
slynnux@slynnux-laptop:~$ echo "hello" | tr -d 'l'
"heo"
```

All of the above input/output redirection operations can also be performed using SSH commands. Let us look at the possibilities:

```
slynnux@gnubox:~$ ssh slynnux-laptop 'cat /etc/passwd | grep root'
slynnux@gnubox:~$ ssh slynnux-laptop 'cat /etc/passwd' > file.txt
slynnux@gnubox:~$ ssh slynnux-laptop 'cat > directed.txt' < file.txt
```

You can also club compression utilities along with SSH:

```
slynnux@gnubox:~$ ssh slynnux-laptop 'tar -czf - file.txt' > file.tar.gz
```

In the above command, we have used `tar -czf` to create a tarball file. `'tar -czf - file.txt'` has - [hyphen] as the file name. When a hyphen is provided as a filename, it implies that the output is not written to a file; instead, it is redirected to standard output.

Now, to list the files in the compressed archive, run the following command:

```
slynnux@gnubox:~$ tar -ztf file.tar.gz
file.txt
```

The SSH protocol also supports data transfer with compression—which comes in handy when bandwidth is an issue. Use the `-C` option with the `ssh` command to enable compression:

```
slynnux@gnubox:~$ ssh -C user@hostname
```

File transfer

SSH also offers the file transfer facility between machines on the network and is highly secure, with SSH being an encrypted protocol. Also, the transfer speed can be improved by enabling compression. Two significant data transfer utilities that use the SSH protocol are SCP and SFTP.

SCP stands for Secure Copy. We can use it to copy files from a local machine to a remote machine, a remote machine to a local machine, and a remote machine to another remote machine.